

Exercise C1

- (a) Prove that the set $S = \{(x_1, x_2, \dots, x_n) : \sum_{i=1}^n a_i x_i \leq b\}$ is convex.
- (b) Prove that if S_1 and S_2 are convex sets, then their intersection $S_1 \cap S_2$ is also convex.
- (c) Deduce that the feasible region of a linear program is convex.

Exercise C2

Ahmed holds two part-time jobs, Job I and Job II. He never wants to work more than a total of 12 hours a week. He has determined that for every hour he works at Job I, he needs 2 hours of preparation time, and for every hour he works at Job II, he needs one hour of preparation time, and he cannot spend more than 16 hours preparing. If he makes \$40 an hour at Job I, and \$30 an hour at Job II how many hours should he work per week at each job to maximize his income?

Exercise C3

Farmer Jones bakes two types of cake (chocolate and vanilla) to supplement his income. Each chocolate cake can be sold for \$1, and each vanilla cake can be sold for 50¢ (\$ 0.5). Each chocolate cake requires 20 minutes of baking time and uses 4 eggs. Each vanilla cake requires 40 minutes of baking time and uses 1 egg. Eight hours of baking time and 30 eggs are available. Formulate an LP to maximize Farmer Jones's revenue, then graphically solve the LP. (A fractional number of cakes is okay.)

Exercise C4

During each 4-hour period, the Algiers police force requires the following number of on-duty police officers: 12 midnight to 4 A.M. $\Rightarrow$ **8** ; 4 to 8 A.M. $\Rightarrow$ **7**; 8 A.M. to 12 noon $\Rightarrow$ **6**; 12 noon to 4 P.M. $\Rightarrow$ **6**; 4 to 8 P.M. $\Rightarrow$ **5**; 8 P.M. to 12 midnight $\Rightarrow$ **4**. Each police officer works two consecutive 4-hour shifts. Formulate an LP that can be used to minimize the number of police officers needed to meet Algiers's daily requirements.

Exercise P1

Consider the following problem, where the value of c_1 is still unknown.

$$\text{Maximize } Z = c_1x_1 + x_2,$$

subject to

$$x_1 + x_2 \leq 6$$

$$x_1 + 2x_2 \leq 10$$

$$x_1 \geq 0, x_2 \geq 0$$

Use graphical analysis to determine the optimal solution(s) for (x_1, x_2) for the various possible values of c_1 ($-\infty < c_1 < \infty$).

Exercise P2 (Optional)

The demand for an item over the next four quarters is 280, 400, 450, and 300 units, respectively. The price per unit starts at \$20 in the first quarter and increases by \$1 each quarter thereafter. The supplier can provide no more than 400 units in any one quarter. Although we can take advantage of lower prices in early quarters, a storage cost of \$3.80 is incurred per unit per quarter. In addition, the number of units that can be held over from one quarter to the next must be 80 or less. Develop an LP model to determine the optimum schedule for purchasing the item to meet the demand

Exercise P3 (Optional)

Consider the following problem, where the values of c_1 and c_2 have not yet been ascertained.

$$\text{Maximize } Z = c_1x_1 + c_2x_2,$$

subject to

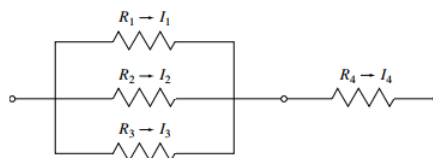
$$2x_1 + x_2 \leq 11$$

$$-x_1 + 2x_2 \leq 2$$

$$x_1 \geq 0, x_2 \geq 0.$$

Use graphical analysis to determine the optimal solution(s) for (x_1, x_2) for the various possible values of c_1 and c_2 . (Hint: Separate the cases where $c_2 = 0$, $c_2 > 0$, and $c_2 < 0$. For the latter two cases, focus on the ratio of c_1 to c_2 .)

Exercise P4 (Optional)



In the electrical circuit of Figure above, I_t = current (in amperes) flowing through resistor t , V_t = voltage drop (in volts) across resistor t , and R_t = resistance (in ohms) of resistor t . Kirchoff's Voltage and Current Laws imply that $V_1 = V_2 = V_3$ and $I_1 + I_2 + I_3 = I_4$. The power dissipated by the current flowing through the resistor $I_t^2 R_t$.

Ohm's Law implies that $V_t = I_t R_t$. The two parts of this problem should be solved independently.

- (a) Suppose you are told that $I_1 = 4$, $I_2 = 6$, $I_3 = 8$, and $I_4 = 18$ are required.

Also, the voltage drop across each resistor must be between 2 and 10 volts. Choose the R_t 's to minimize the total dissipated power. Formulate an LP whose solution will solve your problem.

- (b) Suppose you are told that $V_1 = 6$, $V_2 = 6$, $V_3 = 6$, and $V_4 = 4$ are required.

Also, the current flowing through each resistor must be between 2 and 6 amperes. Choose the R_t 's to minimize the total dissipated power. Formulate an LP whose solution will solve your problem. (Hint: Let $\frac{1}{R_t}$ ($t = 1, 2, 3, 4$) be your decision variables.)